

SMART-SIP+ Business Case Report

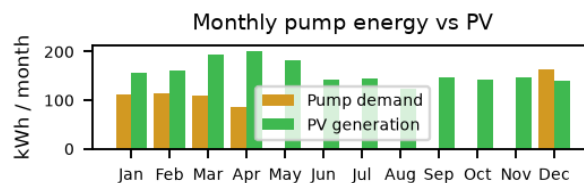
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1. Site, crop and climate

District / location	Comilla (23.9690, 91.0524)
Crop / planting month	Boro Rice / Dec (150-day season)
Area	0.813 ha (2.01 acres / 6.1 bigha)
Climate source	Google Earth Engine - ECMWF ERA5-Land (cell ~9 km at the parcel centroid)
Mean GHI / mean cell temperature	4.49 kWh/m ² /day (x1.00 calibration) / 47.9 C
Irrigation efficiency	0.60

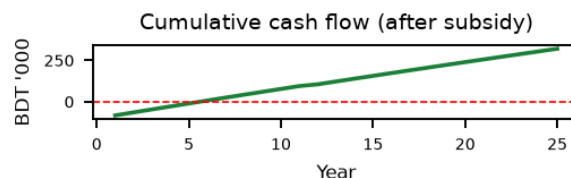
2. Water and hydraulic design

Season irrigation depth / volume	1,667 mm / 13,552 m ³
Peak daily demand / design flow	100 m ³ /day / 12.5 m ³ /h
Static level + drawdown + friction + 1 m	4.5 + 2.0 + 2.0 + 1 = 9.5 m TDH
Pump / PV array	0.54 kW / 1.42 kWp
Share of pumping energy from solar	96%
Land preparation	about 16 pumping days



3. Financial results

CAPEX (gross / after subsidy)	BDT 145,420 / BDT 101,794 (30% subsidy)
Annual benefits	diesel BDT 17,307; cold BDT 0; e-rickshaw BDT 0; grid BDT 3,174
Annual O&M; + grid import	BDT 2,181
Net annual saving / simple payback	BDT 18,299 / 5.6 years
NPV 25 yr @ 8% / IRR	BDT 82,265 / 16.9%
LCOE solar (gross / subsidised)	BDT 8.9 / 6.6 per kWh; diesel fuel only BDT 30.8
Loan	none



4. Environment and groundwater screen

Diesel displaced / CO ₂ avoided	157 L/yr / 0.42 t/yr
Groundwater screen	Medium (score 2) - extraction ratio 4.1x, lift 6.5 m

• Irrigation depth 1,667 mm/yr is 4.1x the assumed recharge (20% of 2,026 mm rain). Screening flag for over-extraction: consider water-saving irrigation or a less thirsty crop.

5. Monthly balance

Month	GHI	ETo	Rain	Irrig mm	Vol m3	Pump kWh	PV kWh	Surplus
Jan	4.3	2.8	2	320	2,602	112	157	45
Feb	4.9	3.7	20	323	2,630	113	161	48
Mar	5.5	4.5	69	310	2,519	109	195	86
Apr	5.9	5.3	94	249	2,026	87	202	114
May	5.2	5.0	403	0	0	0	183	183
Jun	4.2	4.2	256	0	0	0	142	142
Jul	4.1	4.1	280	0	0	0	144	144
Aug	3.5	3.7	351	0	0	0	123	123
Sep	4.3	3.8	255	0	0	0	148	148
Oct	4.0	3.4	212	0	0	0	142	142
Nov	4.2	3.1	54	0	0	0	147	147
Dec	3.8	2.7	30	464	3,775	163	140	0

6. Components

Component	Specification	Total (BDT)
Solar Panels	1.42 kWp (4 x 450 W)	120,530
Pump Inverter (MPPT)	0.54 kW VFD drive	6,456
Mounting Structure	Ground-mount galvanised steel	11,344
Installation & Wiring	Labour, DC/AC cabling, commissioning	7,090
TOTAL (before subsidy)		145,420
After 30% Subsidy		101,794

Data status. Unit costs, prices, cold-storage and e-rickshaw parameters are placeholder assumptions to be replaced by supplier quotes. ERA5-Land is a ~9 km reanalysis: calibrate GHI against local measurements where possible. The groundwater screen is an indicator, not a hydrogeological assessment. Hargreaves-Samani ETo can run high in humid monsoon months.